

As $t \rightarrow \infty$ we find

$$\overline{x \dot{x}} = \frac{1}{2} \frac{I \tau^2}{m} \quad (181)$$

which is a constant. This correlation results from the fact that the particle velocity during the last stopping time τ is related to its present velocity. However, since $\overline{x^2}$ becomes large with t and $\overline{x \dot{x}}$ remains bounded, we see that for large times there is essentially no correlation between x and $\dot{x}$.

The Time Development of an Initial Distribution

From Eq. (154) and (171) for $\dot{x}$ and $\bar{x}^2$ and by making use of the central limit theorem we can write down the probability that a particle has velocity $\dot{x}$ at time t given that it has velocity $\dot{x}_0$ at $t = 0$. From the central limit theorem we have for $P(\dot{x})d\dot{x}$

$$P(\dot{x})d\dot{x} = \frac{1}{\sqrt{\frac{2\pi(\dot{x} - \bar{\dot{x}})^2}{2(\dot{x} - \bar{\dot{x}})^2}}} \exp \left\{ -\frac{(\dot{x} - \bar{\dot{x}})^2}{2(\dot{x} - \bar{\dot{x}})^2} \right\} d\dot{x} \quad (182)$$

Substituting from (154) and (171) we get

$$P(\dot{x})d\dot{x} = \frac{1}{\sqrt{\frac{2\pi kT}{m}(1 - e^{-2t/\tau})}} \exp \left\{ -\frac{(\dot{x} - \dot{x}_0 e^{-t/\tau})^2}{\frac{2kT}{m}(1 - e^{-2t/\tau})} \right\} d\dot{x} \quad (183)$$

In the same way we can obtain the probability that we find a particle at x and dx given that it started at x_0 with velocity $\dot{x}_0$ at time $t = 0$. From (175) and (177) for $\bar{x}$ and $\bar{x}^2$ we get

$$P(x)dx = \frac{1}{\sqrt{\frac{4\pi kT}{m} \tau \left[t - 2\tau(1 - e^{-t/\tau}) + \frac{\tau}{2}(1 - e^{-2t/\tau}) \right]}} \exp \left\{ -\frac{\left[x - x_0 - \dot{x}_0 \tau (1 - e^{-t/\tau}) \right]^2}{\frac{4kT}{m} \tau \left[t - 2\tau(1 - e^{-t/\tau}) + \frac{\tau}{2}(1 - e^{-2t/\tau}) \right]} \right\} \quad (184)$$

We may now write down a formula for the time development of the velocity distribution. We have in general

$$P(\dot{x}, t) d\dot{x} = \int P_0(\dot{x}_0) P(\dot{x}_0 | \dot{x} t) d\dot{x}_0 \quad (185)$$

$$P(\dot{x}, t) d\dot{x} = \int \frac{P_0(\dot{x}_0)}{\sqrt{\frac{2\pi kT}{m} (1 - e^{-2t/\tau})}} \exp \left\{ - \frac{(\dot{x} - \dot{x}_0 e^{-t/\tau})^2}{\frac{2kT}{m} (1 - e^{-2t/\tau})} \right\} d\dot{x}_0 \quad (186)$$

As t goes to infinity, $P(\dot{x} t | \dot{x}_0)$ becomes independent of $\dot{x}_0$ and (184) can be written as

$$\begin{aligned} P(\dot{x}, t) &\xrightarrow{t \rightarrow \infty} \frac{1}{\sqrt{\frac{2\pi kT}{m}}} \exp \left\{ - \frac{m\dot{x}^2}{2kT} \right\} \int P_0(\dot{x}_0) d\dot{x}_0 \\ &= \frac{1}{\sqrt{\frac{2\pi kT}{m}}} \exp \left\{ - \frac{m\dot{x}^2}{2kT} \right\} \quad (186) \end{aligned}$$

Thus all distributions go to Maxwellian for large t .

In a similar manner we can find the probability that a particle is at x given that at $t = 0$ the probability that it had velocity $\dot{x}_0$ and position x_0 was

$$P_0(\dot{x}_0, x_0) d\dot{x}_0 dx_0$$

$$P(x, t) dx = dx \iint P(x_0, \dot{x}_0 | x, t) P_0(x_0, \dot{x}_0) dx_0 d\dot{x}_0 \quad (187)$$

$$= dx \iint dx_0 d\dot{x}_0 P_0(x_0, \dot{x}_0) \exp \left\{ - \frac{[x - x_0 - \dot{x}_0 \tau (1 - e^{-t/\tau})]^2}{\frac{4kT\tau}{m} (t - 2\tau(1 - e^{-t/\tau}) + \frac{\tau}{2} (1 - e^{-2t/\tau}))} \right\} \cdot \frac{1}{\left[\frac{4kT\tau}{m} (t - 2\tau(1 - e^{-t/\tau}) + \frac{\tau}{2} (1 - e^{-2t/\tau})) \right]^{1/2}} \quad (188)$$

If t is large compared to τ , (188) can be approximated by

$$P(x) dx = dx \iint P(x_0, \dot{x}_0) \exp \left\{ - \frac{[x - x_0 - \dot{x}_0 \tau]^2}{\frac{4kT\tau t}{m}} \right\} dx_0 d\dot{x}_0 \cdot \frac{1}{\left[\frac{4kT\tau t}{m} \right]^{1/2}} \quad (189)$$

Further if $\dot{x}_0 \tau$ is small as it usually is, (189) can be further reduced to

$$P(x) dx = dx \int \frac{P(x_0) dx_0}{\left[\frac{4kT\tau t}{m} \right]^{1/2}} \exp \left\{ - \frac{[x - x_0]^2}{\frac{4kT\tau t}{m}} \right\} \quad (190)$$

Relation to the Diffusion Equation

The equation for the diffusion of a substance through a uniform unbounded medium is

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2} \quad (191)$$

First we claim that a solution of this equation is

$$n(x,t) = \sqrt{\frac{1}{4Dt}} e^{-\frac{x^2}{4Dt}} \quad (192)$$

Proof.

$$\frac{\partial n}{\partial t} = \sqrt{\frac{1}{4Dt}} \left\{ -\frac{1}{2} t^{-3/2} + \frac{x^2}{4Dt^5/2} \right\} e^{-\frac{x^2}{4Dt}} \quad (193)$$

$$\frac{\partial n}{\partial x} = \sqrt{\frac{1}{4Dt}} \left(-\frac{x}{2Dt} e^{-\frac{x^2}{4Dt}} \right) \quad (194)$$

$$\frac{\partial^2 n}{\partial x^2} = \sqrt{\frac{1}{4Dt}} \left\{ -\frac{1}{2Dt^3/2} + \frac{x^2}{4Dt^5/2} \right\} e^{-\frac{x^2}{4Dt}} \quad (195)$$

Multiplying (195) by D we see we get (193) and hence (191) is satisfied.

The solution (192) has the property that its integral is 1. Since the equation is linear any multiple of (192) is also a solution. If (192) is multiplied by n_0 the solution represents what would happen if we started out with n_0 particles at $x = 0$ at $t = 0$. If we had started with n particles at x_0 at t equal 0 the solution would be

$$n(x,t) = n \sqrt{\frac{1}{4Dt}} e^{-\frac{(x-x_0)^2}{4Dt}} \quad (196)$$

Again since the equation is linear, the sum of solutions is also a solution.

Thus, if at $t = 0$ we have n_i particles at points x_i , the solution would have been

$$n(x,t) = \sum_i n_i \sqrt{\frac{1}{4\pi Dt}} e^{-\frac{(x-x_i)^2}{4Dt}} \quad (197)$$

If initially we have a continuous distribution of particles, then we must replace the sum in (197) by an integral.

$$n(x,t) = \int n(x_0) \sqrt{\frac{1}{4\pi Dt}} e^{-\frac{(x-x_0)^2}{4Dt}} dx_0 \quad (198)$$

We see that Eq. (198) is exactly the same form as (190). Thus for times long compared to τ the spatial distribution of Brownian particles is determined by a diffusion equation with the diffusion constant D being given by

$$D = \frac{kT}{m} \tau = V_T^2 \tau \quad (190)$$

For spherical particles of radius R in a fluid with viscosity η , τ is given by

$$\tau = \frac{m}{6\pi R \eta} \quad (191)$$

Diffusion in Velocity Space

From Eq. (186) for $P(\dot{x})$ we see that for times short compared to τ , $P(\dot{x})$ can be approximated by

$$P(\dot{x},t) d\dot{x} \approx \int \frac{P(\dot{x}_0)}{\sqrt{\frac{4\pi V_T^2 t}{\tau}}} \exp \left\{ -\frac{(\dot{x}-\dot{x}_0)^2}{\frac{4V_T^2 t}{\tau}} \right\} d\dot{x}_0 \quad (192)$$

This is also identical to Eq. (198) and so for short times the distribution of velocities is determined by a diffusion equation with D given by

$$D = v_T^2 / \tau$$

The Maxwell distribution results from the competition between this tendency to diffuse in velocity space and the systematic drag on a particle which tends to slow it down.

Example of Brownian Motion

Consider a particle with radius 10^{-4} , with a density of 2 gm/cm^3 suspended in water

$$m = 10^{-12} \text{ gm}$$

$$\eta = 10^{-2} \frac{\text{dyne sec}}{\text{cm}^2}$$

$$\tau = \frac{8}{18} \times 10^{-6} \text{ sec} \approx 5 \times 10^{-7} \text{ sec}$$

at 300°K

$$kT = 4 \times 10^{-14} \text{ ergs}$$

$$v^2 = 10^{-2} \text{ cm}^2/\text{sec}^2$$

$$v\tau = .444 \times 10^{-7} \text{ cm}$$

$$D = v^2 \tau = 5 \times 10^{-9} \text{ cm}^2/\text{sec}$$

The Fokker Planck Equation

For a Brownian particle we saw that the velocity was determined by two effects, the systematic drag and the diffusion resulting from the random bombardment. We may generalize the results we found there to

The Relation Between the Spectral Density and Diffusion

We have found that for large times the average value of x^2 is given

by

$$\overline{\Delta x^2} = \overline{x^2} - \frac{2}{x} = \frac{\tau^2 I}{m^2}$$

$$\frac{I}{m^2} = \int_{-\infty}^{\infty} \overline{a(t') a(t' - \xi)} d\xi .$$

Let $t' - \xi = t$

$$t' = t + \xi$$

$$\frac{I}{m^2} = \int_{-\infty}^{\infty} \overline{a(t) a(t + \xi)} d\xi .$$

For a stationary random process $\overline{a(t) a(t + \xi)} = c(\xi)$

$$\frac{I}{m^2} = \int_{-\infty}^{\infty} c(\xi) dx \cong \int_{-T/2}^{T/2} c(\xi) d\xi \quad T \text{ is large .}$$

Making use of the relation between the spectral density and the correlation function II. 7

$$\begin{aligned} \frac{I}{m^2} &= \int_{-T/2}^{T/2} \int_0^{\infty} G(\omega) \cos \omega x dx d\omega \\ &= \int_0^{\infty} G(\omega) \frac{2 \sin \omega T/2}{\omega} d\omega = \pi G(0) . \end{aligned}$$

Thus the diffusion in x space is controlled by the spectral density at zero frequency.

encompasses any process which consists of a systematic variation plus a diffusion.

Let x be a variable of interest and let the conditional probability that x lies between x and $x + dx$ at times t given that it had the value of x_0 at time $t = 0$ be

$$P(x_0 | x, t) dx . \quad (193)$$

For a Markoff process the probability that x lies between x and $x + dx$ at $t + \Delta t$ is

$$P(x_0 | x, t + \Delta t) = \int P(x_0 | y, t) P(y | x, \Delta t) dy . \quad (194)$$

Consider now the average value of a function $R(x)$

$$\bar{R} = \int R(x) P(x_0 | x, t) dx = \overline{R(t)} . \quad (195)$$

Taking the time derivative of $\bar{R}$ gives

$$\begin{aligned} \frac{d\bar{R}}{dt} &= \int R(x) \frac{\partial P(x_0 | x, t)}{\partial t} dx = \\ &= \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int R(x) \left\{ P(x_0 | x, t + \Delta t) - P(x_0 | x, t) \right\} dx \\ &= \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int R(x) dx \left\{ \int P(x_0 | y, t) P(y | x, \Delta t) dy - P(x_0 | x, t) \right\} . \end{aligned} \quad (196)$$

Interchanging the order of integration in the double integral gives

$$\int dy P(x_0 | y, t) \int R(x) P(y | x, \Delta t) dx . \quad (197)$$

Now if Δt is small, $P(y | x, \Delta t)$ must be small for x 's which differ appreciably from y . Therefore, we expand $R(x)$ about $x = y$.

$$R(x) = R(y) + R'(y)(x-y) + \frac{R''(y)}{2} (x-y)^2 + \dots \quad (198)$$

$$\begin{aligned} \int dx R(x) P(y|x, \Delta t) &= \int \left(P(y|x, \Delta t) \left(R(y) + R'(y)(x-y) + \frac{R''(y)}{2} (x-y)^2 \right) \right) dx \\ &= R(y) + R'(y) \overline{(x-y)}_{\Delta t} + \frac{R''(y)}{2} \overline{(x-y)^2}_{\Delta t} \end{aligned} \quad (199)$$

Now

$$\overline{(x-y)}_{\Delta t}$$

is the average value of the displacement of the mean of the distribution during the time Δt and we take it to be proportional to Δt .

$$\overline{(x-y)}_{\Delta t} = A(y)\Delta t \quad (200)$$

This gives the systematic development of y .

Likewise

$$\overline{(x-y)^2}_{\Delta t}$$

is the second moment of the distribution about its mean at time Δt (to order Δt). It is proportional to the square of the width of the distribution for a Gaussian. For diffusion, as we have seen it, is proportional to Δt .

Thus we assume it is proportional to Δt and write

$$\overline{(x-y)^2}_{\Delta t} = B(y)\Delta t \quad (201)$$

This is the diffusion term for the process.

Substituting in $\frac{d\bar{R}}{dt}$ gives

$$\frac{d\bar{R}}{dt} = \int dy R(y) \frac{\partial P(x_o | yt)}{\partial t} = \int \left(A(y) \cdot \frac{\partial R}{\partial y} + \frac{B(y)}{2} \frac{\partial^2 R}{\partial y^2} \right) P(x_o | yt) dy . \quad (202)$$

Integrating the A term by parts with respect to y gives

$$\begin{aligned} \int A(y) \frac{\partial R}{\partial y} P(x_o | yt) dy &= A(y) P(x_o | yt) R(y) \Big|_{-\infty}^{\infty} \\ &\quad - \int R(y) \frac{\partial}{\partial y} A(y) P(x_o | yt) dy \\ &= - \int R(y) \frac{\partial}{\partial y} A(y) P(x_o | yt) dy . \end{aligned} \quad (203)$$

Likewise integrating the B term by parts twice gives

$$\int \frac{B(y)}{2} \frac{\partial R}{\partial y^2} P(x_o | yt) dy = \int \frac{R(y)}{2} \frac{\partial^2}{\partial y^2} [B(y) P(x_o | yt)] dy . \quad (204)$$

Finally (202) may be written

$$\int \left\{ \frac{\partial P(x_o | yt)}{\partial t} + \frac{\partial A(y) P(x_o | yt)}{\partial y} - \frac{1}{2} \frac{\partial^2 B(y) P(x_o | yt)}{\partial y^2} \right\} R(y) dy . \quad (205)$$

Now since $R(y)$ was an arbitrary function the only way to satisfy this equation for all R 's is to have the term in $\{ \}$ equal to zero.

$$\frac{\partial P(x_o | yt)}{\partial t} + \frac{\partial}{\partial y} A(y) P(x_o | yt) - \frac{1}{2} \frac{\partial^2}{\partial y^2} B(y) P(x_o | yt) = 0 . \quad (206)$$

We may multiply this by an arbitrary distribution in x_o at time 0

$$P(x_o)$$

and integrate over all x_o to obtain an equation for $P(yt)$,

$$\frac{\partial P(y,t)}{\partial t} + \frac{\partial}{\partial y} A(y)P(y,t) - \frac{1}{2} \frac{\partial^2}{\partial y^2} B(y)P(y,t) \quad (207)$$

This is the Fokker Planck equation.

The Fokker Planck Equation in N Dimension

Let $\vec{r}$ be an N dimensional vector which describes the state of a system. Let

$$P(\vec{r}_0 | \vec{r}, t) d^N r, \quad \vec{r} = \sum_i \vec{e}_i x_i, \quad d^N r = dx_1 \dots dx_N \quad (208)$$

be the probability that the system is in $d^N r$ at $\vec{r}$ at time t given that it was at $\vec{r}_0$ at $t = 0$. Let $R(\vec{r})$ be an arbitrary function of $\vec{r}$. The average value of R is given by

$$\overline{R(t)} = \int R(\vec{r}) P(\vec{r}_0 | \vec{r}, t) d^N r \quad (209)$$

By the definition of the time derivative we have

$$\begin{aligned} \frac{d\overline{R(t)}}{dt} &= \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int R(\vec{r}) \left(P(\vec{r}_0 | \vec{r}t + \Delta t) - P(\vec{r}_0 | \vec{r}t) \right) d^N r \\ &= \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int R(\vec{r}) \left\{ \int P(\vec{r}_0 | \vec{r}^1 t) P(\vec{r}^1 | \vec{r} \Delta t) d^N r^1 - P(\vec{r}_0 | \vec{r}t) \right\} d^N r \end{aligned} \quad (210)$$

where again we have assumed that

$$P(\vec{r}^1 t | \vec{r}t + \Delta t) = P(\vec{r}^1 | \vec{r} \Delta t)$$

Again interchanging the order of the $\vec{r}$ and $\vec{r}^1$ integration assuming that $P(\vec{r}^1 | \vec{r} \Delta t)$ is small for $|\vec{r} - \vec{r}^1|$ large and expanding $R(\vec{r})$ about $R(\vec{r}^1)$ for the double integral on the left hand side of (210) ,

$$\begin{aligned}
 & \int d^N r^1 P(\vec{r}_0 | \vec{r}^1, t) \int d^N r P(\vec{r}^1 | \vec{r}, \Delta t) \left\{ R(\vec{r}^1) + \sum_i \left(\frac{\partial R}{\partial x_i} \right)_{\vec{r}^1} (x_i - x_i^1) \right. \\
 & \quad \left. + \frac{1}{2} \sum_{ij} \left(\frac{\partial^2 R}{\partial x_i \partial x_j} \right)_{\vec{r}^1} (x_i - x_i^1)(x_j - x_j^1) \right. \\
 & \quad \left. \dots \right\} \\
 & \approx \int d^N r^1 P(\vec{r}_0 | \vec{r}^1, t) \left\{ R(\vec{r}^1) + \sum_i A_i(\vec{r}^1) \Delta t \frac{\partial R}{\partial x_i^1} + \frac{1}{2} \sum_{ij} B_{ij}(\vec{r}^1) \Delta t \frac{\partial^2 R}{\partial x_i^1 \partial x_j^1} \right\} \quad (211)
 \end{aligned}$$

where

$$A_i(\vec{r}^1) \Delta t = \int (x_i - x_i^1) P(\vec{r}^1 | \vec{r} \Delta t) d^N r \quad (212)$$

$$B_{ij}(\vec{r}^1) \Delta t = \int (x_i - x_i^1)(x_j - x_j^1) P(\vec{r}^1 | \vec{r} \Delta t) d^N r \quad (213)$$

Integrating

$$\int P(\vec{r}_0 | \vec{r}^1, t) \frac{\partial R}{\partial x_i^1} A_i(\vec{r}^1) d^N r^1 \quad (214)$$

by parts with respect to x_i^1 give

$$\begin{aligned}
 & \int P(\vec{r}_0 | \vec{r}^1, t) A_i(\vec{r}^1) \frac{\partial R}{\partial x_i^1} d^N r^1 = \int P(\vec{r}_0 | \vec{r}^1, t) A_i(\vec{r}^1) R(\vec{r}^1) \Big|_{x_i^1 = -\infty}^{x_i^1 = \infty} d^{N-1} r^{11} \\
 & \quad - \int R(\vec{r}^1) \frac{\partial}{\partial x_i^1} A_i(\vec{r}^1) P(\vec{r}_0 | \vec{r}^1, t) d^N r^1 \\
 & = - \int R(\vec{r}^1) \frac{\partial}{\partial x_i^1} A_i(\vec{r}^1) P(\vec{r}_0 | \vec{r}^1, t) d^N r^1 \quad (215)
 \end{aligned}$$

where $d^{N-1} r^{11}$ means that we integrate over all variables except x_i^1 . P

should vanish sufficiently rapidly to make this integral over r'' vanish. Likewise integrating

$$\int P(\vec{r}_o | \vec{r}^1, t) B_{ij}(\vec{r}^1) \frac{\partial^2 R}{\partial x'_i \partial x'_j} d^N r^1 \quad (216)$$

gives

$$\int R(\vec{r}^1) \left\{ \frac{\partial^2}{\partial x'_i \partial x'_j} P(\vec{r}_o | \vec{r}^1, t) B_{ij}(\vec{r}^1) \right\} d^N r^1 \quad (217)$$

Substituting (215) and (217) into (211) in (210), gives

$$\begin{aligned} \int R(\vec{r}) \left\{ \frac{\partial P(\vec{r}_o | \vec{r}^1, t)}{\partial t} + \sum_i \frac{\partial}{\partial x'_i} A_i(\vec{r}^1) P(\vec{r}_o | \vec{r}^1, t) \right. \\ \left. - \frac{1}{2} \sum_{ij} \frac{\partial}{\partial x'_i} \frac{\partial}{\partial x'_j} B_{ij} P(\vec{r}_o | \vec{r}^1, t) \right\} d^N r^1 = 0 \quad (218) \end{aligned}$$

Since $R(\vec{r}^1)$ is an arbitrary function the only way this can be satisfied

is if the expression in { } is zero. Thus we get the general Fokker

Planck equation

$$\begin{aligned} \frac{\partial P(\vec{r}_o | \vec{r}t)}{\partial t} + \sum_i \frac{\partial}{\partial x'_i} A_i(\vec{r}) P(\vec{r}_o | \vec{r}t) \\ - \frac{1}{2} \sum_{ij} \frac{\partial}{\partial x'_i} \frac{\partial}{\partial x'_j} B_{ij}(\vec{r}) P(\vec{r}_o | \vec{r}t) = 0 \quad (219) \end{aligned}$$

or by multiplying by an arbitrary initial distribution of $\vec{r}_o$ and integrating over $\vec{r}_o$

$$\frac{\partial P(\vec{r}^1, t)}{\partial t} + \sum_i \frac{\partial}{\partial x'_i} A_i(\vec{r}) P(\vec{r}t) - \frac{1}{2} \sum_{ij} \frac{\partial}{\partial x'_i} \frac{\partial}{\partial x'_j} B_{ij}(\vec{r}) P(\vec{r}t) = 0 \quad (220)$$

Problem

Obtain the Fokker Planck Equation for the diffusion of a Brownian Particle in 2 dimensions for both cartesian and polar coordinates. Assume the drags is isotropic. Do the same in 3 dimensions for Cartesian, cylindrical and spherical coordinates.

We may apply the Fokker Planck equation to the problem of Brownian motion. For this case the velocity satisfies the Langevinequation. We have

$$\frac{dv}{dt} = -\frac{v}{\tau} + a(t) \quad (221)$$

$$\Delta v = -\frac{v\Delta t}{\tau} + \int_0^{\Delta t} a(t)dt \quad (222)$$

$$\overline{\frac{\Delta v}{\Delta t}} = -\frac{v}{\tau} = A(v) \quad (223)$$

$$\overline{\Delta v^2} = \overline{\frac{v^2 \Delta t^2}{\tau^2} - \frac{2v\Delta t}{\tau} \int_0^{\Delta t} a(t)dt + \int_0^{\Delta t} \int_0^{\Delta t} a(t_1)a(t_2)dt_1 dt_2} \quad (224)$$

$$\overline{\Delta v^2} = \frac{I}{m} \frac{\Delta t}{2} \quad (225)$$

$$\overline{\frac{\Delta v^2}{\Delta t}} = \frac{I}{m} \frac{1}{2} = \frac{2kT}{m\tau} = B(v) \quad (226)$$

The Fokker Planck equation for the velocity is

$$\frac{\partial P(v,t)}{\partial t} - \frac{\partial}{\partial v} \frac{vP(v,t)}{\tau} - \frac{1}{2} \frac{\partial^2}{\partial v^2} \frac{2kT}{m\tau} P(v,t) = 0. \quad (227)$$

We can readily verify that a Maxwellian is a time independent solution of

this equation. Setting P equal to $\exp\left(-\frac{mv^2}{2kT}\right)$ and substituting in the above equation gives

$$\frac{\partial}{\partial v} \left\{ \left[\frac{v}{\tau} P(v) \right] + \frac{kT}{m\tau} \left(\frac{-mv}{kT} \right) P(v) \right\} = 0. \quad (228)$$

We should also like to find the general time dependent solution to this equation for an arbitrary initial distribution. We have of course already found this in Eq. (189) and we could simply substitute this expression in (227) and verify that it is a solution. We shall, however, solve (227) by a different method since some useful mathematical methods are involved.

First we Fourier analyze P and write

$$P(v,t) = \frac{1}{\sqrt{2\pi}} \int f(k,t) e^{ikv} dk \quad (229)$$

$$f(k,t) = \frac{1}{\sqrt{2\pi}} \int P(v) e^{-ikv} dv \quad (230)$$

Substituting (229) in (227) gives

$$\int \left\{ \frac{\partial f}{\partial t} - \frac{ikv}{\tau} f - \frac{f}{\tau} + \frac{kT}{m\tau} k^2 f \right\} e^{ikv} dk = 0 \quad (231)$$

Consider the term

$$- \int_{-\infty}^{\infty} \frac{ikv}{\tau} f e^{ikv} dk \quad (232)$$

Integrating by parts gives

$$\begin{aligned}
 & - \frac{kfe^{ikv}}{\tau} \Big|_{-\infty}^{\infty} + \frac{1}{\tau} \int_{-\infty}^{\infty} e^{ikv} (f+kf^1) dk \\
 & = \frac{1}{\tau} \int_{-\infty}^{\infty} e^{ikv} (f+kf^1) dk \quad . \quad (233)
 \end{aligned}$$

Thus equation (231) becomes

$$\int \left\{ \frac{\partial f}{\partial t} + \frac{kf^1}{\tau} + \frac{v^2 k^2}{\tau} f \right\} e^{ikv} dk = 0 \quad . \quad (234)$$

For this equation to be satisfied the expression in brackets must be zero for all k . You can verify this by multiplying by $e^{-ik^1 v}$ and integrating over all v . Thus we have

$$\frac{\partial f}{\partial t} + \frac{kf^1}{\tau} + \frac{v^2 k^2}{\tau} f = 0 \quad . \quad (235)$$

This differential equation is of the form that can be solved by means of the methods of characteristics. In general we have for df

$$df = \left(\frac{\partial f}{\partial t} \right)_k dt + \left(\frac{\partial f}{\partial k} \right)_t dk \quad . \quad (236)$$

Suppose we ask for the change in f along some curve in the k, t plane

$$t = t(k) \quad .$$

Then dt is related to dk by

$$dt = \frac{dt}{dk} dk \quad (237)$$

and we have

$$df = \left(\frac{\partial f}{\partial t} \right)_k \cdot \frac{dt}{dk} dk + \left(\frac{\partial f}{\partial k} \right)_t \cdot dk \quad , \quad (238)$$

or

$$\left(\frac{df}{dk} \right)_c = \left(\frac{\partial f}{\partial t} \right)_k \frac{dt}{dk} + \left(\frac{\partial f}{\partial k} \right)_t \quad (239)$$

We see that if

$$\frac{dt}{dk} = \frac{\tau}{k} \quad (240)$$

that the first two terms of (235) can be written in the form

$$\frac{k}{\tau} \left(\frac{df}{dk} \right)_c \quad (241)$$

where c is determined by (237).

Eq. (237) can be integrated to give

$$\ln \frac{k}{k_0} = \frac{t}{\tau} \quad (242)$$

or

$$k = k_0 e^{t/\tau} \quad (243)$$

This is a family of curves in the k, t plane. We may now write Eq. (235)

in the form

$$\frac{k}{\tau} \left(\frac{df}{dk} \right)_c + \frac{v_T^2 k^2}{\tau} f = 0 \quad (244)$$

or

$$\ln \frac{f(k)}{\psi(k_0)} = - \frac{v_T^2 k^2}{2} \quad (246)$$

$$f_c(k) = \psi(k_0) e^{- \frac{v_T^2 k^2}{2}} \quad (247)$$

According to (243) at $t = 0, k = k_0$, thus

$$f_c(k_o) = \psi(k_o) e^{-\frac{v^2 k_o^2}{2T}} \quad (248)$$

Thus

$$\psi(k_o) = f(k_o) e^{\frac{v^2 k_o^2}{2T}} \quad (249)$$

$$f_1(k) = f(k_o) e^{\frac{v^2}{2T} (k_o^2 - k^2)} \quad (250)$$

$$f(k,t) = f_o(ke^{-t/\tau}) e^{\frac{v^2 k^2}{2T} (e^{-t/\tau} - 1)} \quad (251)$$

If at $t = 0$, $P(v,0)$ is a delta function

$$P(v,0) = \delta(v - v_o) \quad (252)$$

then $f(k,0)$ is given by

$$f(k,0) = \frac{1}{\sqrt{2\pi}} e^{-ikv_o} \quad (253)$$

$$f(k,t) = \frac{1}{\sqrt{2\pi}} \left[e^{-iv_o k e^{-t/\tau}} + \frac{v^2 k^2}{2T} (e^{-t/\tau} - 1) \right] \quad (254)$$

$$\begin{aligned} f(v,t) &= \frac{1}{2\pi} \int dk \left[\exp \left\{ -\frac{k^2 v^2}{2T} (1 - e^{-t/\tau}) + ik(v - v_o e^{-t/\tau}) \right\} \right] \\ &= \frac{1}{2\pi} \int dk \exp \left\{ -\frac{v^2 (1 - e^{-t/\tau})}{2} \left[k + \frac{1(v - v_o e^{-t/\tau})}{v^2 (1 - e^{-t/\tau})} \right]^2 \right. \\ &\quad \left. - \frac{(v - v_o e^{-t/\tau})^2}{2v^2 (1 - e^{-t/\tau})} \right\} \end{aligned} \quad (255)$$

$$f(v,t) = e^{-\left[\frac{(v-v_0 e^{-t/\tau})^2}{2v_T^2(1-e^{-t/\tau})} \right]} \quad (256)$$

Problem II.

Find the Fokker Planck equation for the RC circuit driven by the shot noise from a vacuum tube. See Problem I.